

Econometric Formulation of the Problem

A hierarchical panel model for predicting the sign of a future return

1. Panel structure

The data form a panel. Allocation $a = 1, \dots, A$ is observed on date $t = 1, \dots, T$ and belongs to group $g(a)$. Let

$$r_{a,t+1} \text{ be the future return, } y_{a,t} = \mathbf{1}\{r_{a,t+1} > 0\}$$

be the continuous outcome and the associated binary target. At date t we observe:

- $x_{a,t}$: allocation-specific information known at prediction time;
- z_t : the common market state, also known at prediction time.

The objective is to separate what is common to every observation from what is specific to a group, an allocation, or a date.

2. General decomposition

A convenient econometric equation is

$$r_{a,t+1} = \underbrace{\mu}_{\text{global mean}} + \underbrace{\theta_{g(a)}}_{\text{group effect}} + \underbrace{\alpha_a}_{\text{allocation effect}} + \underbrace{z_t^\top \kappa}_{\text{market-state effect}} + \underbrace{x_{a,t}^\top \beta}_{\text{common slopes}} + \underbrace{x_{a,t}^\top b_a}_{\text{slope deviation}} + \underbrace{\varepsilon_{a,t+1}}_{\text{future shock}}. \quad (1)$$

The components have the following meaning:

Term	Interpretation
μ	average return across the whole panel;
$\theta_{g(a)}$	persistent difference between groups;
α_a	allocation a 's deviation from its group;
$z_t^\top \kappa$	effect of the market state shared on date t ;
$x_{a,t}^\top \beta$	response to predictors common to all allocations;
$x_{a,t}^\top b_a$	allocation-specific deviation from the common response;
$\varepsilon_{a,t+1}$	future information not known at date t .

Hence allocation a has total intercept

$$\mu + \theta_{g(a)} + \alpha_a$$

and total slope vector

$$\beta + b_a.$$

This makes the decomposition easy to read: there is a common model, to which we add progressively finer group- and allocation-level corrections.

3. The date effect and forecasting

For an in-sample panel regression one could write an unrestricted date fixed effect λ_t . It is useful conceptually to decompose it as

$$\lambda_t = z_t^\top \kappa + \nu_t,$$

where $z_t^\top \kappa$ is explained by information observed on date t and ν_t is an unpredictable common shock. An unrestricted λ_t cannot be used to forecast a new date because its coefficient is not yet known. Therefore the forecasting equation uses the observable component $z_t^\top \kappa$; the remaining ν_t is absorbed into the forecast error.

4. From return prediction to sign prediction

Define the systematic score, excluding the future shock, by

$$\eta_{a,t} = \mu + \theta_{g(a)} + \alpha_a + z_t^\top \kappa + x_{a,t}^\top \beta + x_{a,t}^\top b_a.$$

There are two equivalent modeling routes.

Continuous route. Estimate Equation (1), predict $\hat{r}_{a,t+1} = \hat{\eta}_{a,t}$, and use

$$\hat{y}_{a,t} = \mathbf{1}\{\hat{\eta}_{a,t} > 0\}.$$

If $\varepsilon_{a,t+1} \sim \mathcal{N}(0, \sigma^2)$, then

$$\Pr(y_{a,t} = 1 \mid x_{a,t}, z_t, a) = \Phi\left(\frac{\eta_{a,t}}{\sigma}\right).$$

Binary route. Specify the sign probability directly as

$$\Pr(y_{a,t} = 1 \mid x_{a,t}, z_t, a) = F(\eta_{a,t}), \tag{2}$$

where F is the logistic cdf (logit) or the standard normal cdf (probit). With either symmetric link, the default 50% decision rule is equivalent to $\hat{\eta}_{a,t} > 0$.

5. Partial pooling and identification

Estimating an unrelated coefficient for every allocation is noisy. A hierarchical specification shares information across the panel:

$$\theta_g \sim \mathcal{N}(0, \sigma_g^2), \quad \alpha_a \sim \mathcal{N}(0, \sigma_a^2), \quad b_a \sim \mathcal{N}(0, \Sigma_b).$$

Small or weakly observed groups and allocations are then shrunk toward the common model, while well-observed ones may differ when the data support it. The same principle can be implemented with penalized fixed effects.

For a pure fixed-effect formulation, impose normalizations such as

$$\sum_g \theta_g = 0, \quad \sum_{a:g(a)=g} \alpha_a = 0, \quad \sum_a b_a = 0,$$

so that the global, group, and allocation components are separately identified.

Core model. Future returns are decomposed into a global component, nested group and allocation effects, an observable market-state effect, common predictor slopes, allocation-specific slope deviations, and an unpredictable shock. The classification target is simply the sign of this latent return.